3 September 2026

Dear Guest Editors and Editors of Fractal and Fractional,

Re: Submission to the Special Issue "Fractional-Order Modeling in Data-Driven Intelligent Systems: From Optimization to AI/ML Applications"

Please consider our manuscript, "Endpoint-Constrained Multi-Band Fractional Memory for Repairing Long-Duration Limb Occlusions in Dance Motion Capture," for publication as an Article in this Special Issue within the Section "Optimization, Big Data, and AI/ML."

The manuscript addresses a specific missing-data problem in markerless motion capture: reconstructing a wrist, ankle, or limb that remains occluded while the rest of a dancer is visible. We introduce a multi-band Grunwald-Letnikov fractional-integral memory bridge that operates only on observed motion increments. Its residual increments are converted into a discrete bridge whose correction is exactly zero at the two observed boundaries of a missing interval. This gives the long-gap model a power-law memory profile without allowing hidden target coordinates to enter the reconstruction.

Previous human-motion studies have used fractional operators as trajectory regularizers. Our work instead uses fractional integral memory to construct inference features from visible increments during partial-body completion. The analytical endpoint constraint and a parameter-matched finite-window control allow this contribution to be tested directly.

The evaluation uses all 1,408 AIST++ sequences, a choreography-grouped split, three training seeds, and a held-out test set of 221 sequences. A focused 72-frame limb-occlusion hypothesis was fixed before test access and evaluated with pre-specified decision rules. The fractional system improved MPJPE over the finite bridge in all three seeds; the paired sequence-bootstrap gain was 0.279% with a 95% confidence interval of 0.167% to 0.387%. A loss-matched follow-up control retained a 0.293% gain after removal of the fractional trajectory loss. The duration analysis also identifies the operating boundary: localized gaps of 60-96 frames benefit, whereas uniform full-body gaps favor finite memory.

The manuscript is directly linked to Fractal and Fractional and the selected Special Issue because the Grunwald-Letnikov fractional integral is the central inference operator, not an auxiliary descriptive statistic. The study integrates a memory-aware fractional model with a learned temporal architecture, provides an analytical endpoint derivation, and evaluates real-time performance on dynamic motion data. The matched control, order ablation, statistical protocol, and GPU timing measurements establish both the fractional contribution and its practical scale.

Protocol definitions, aggregate results, and source code for training, evaluation, statistical analysis, and figure generation are publicly available at https://github.com/liangdagongjue-code/StreetStyle/tree/main/fractional_motion_repair. The repository does not redistribute AIST++ data or trained checkpoints.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. Both authors have approved the manuscript and agree with its submission. The authors declare no conflict of interest.

Thank you for your consideration.

Sincerely,

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